Practice 1: Class and Data Encapsulation Exercise

Define a Warrior class that includes three private attributes: String name, int life, and int magic. It should also include a method void NewMoon() which represents an attack method (New Moon Sword Technique). The attack target can be a warrior or a witch. If the target is a warrior, it deducts 25 life points; if the target is a witch, it deducts 40 life points. Each attack consumes 10 magic points. If the magic points are less than 10, the attack is invalid, and a message is printed. If the attacked character's life points are <= 0, print the name of the deceased character and a death message. Initial life points are 400 and initial magic points are 100.

Define a Witch class that includes three private attributes: String name, int life, and int magic. It should also include a method void SmallFire() which represents a magical attack method (Small Fireball). The attack target can be a warrior or a witch. If the target is a warrior, it deducts 40 life points; if the target is a witch, it deducts 60 life points. Each attack consumes 25 magic points. If the magic points are less than 25, the spell is invalid, and a message is printed. If the attacked character's life points are <= 0, print the name of the deceased character and a death message. Initial life points are 280 and initial magic points are 200.

Create an array of 3 Warrior objects and an array of 3 Witch objects. Repeat the following steps until a character dies:

1. Randomly select a warrior and a witch, then the warrior uses the New Moon Sword Technique to attack the witch once.
2. Randomly select a witch and a warrior, then the witch uses the Small Fireball to attack the warrior once.

Note: Use Math.random() to generate a decimal number between 0 and 1.